PRODUCT REQUIREMENTS DOCUMENT

CRIME SIGHT AI

An AI-powered crime analytics and intelligence platform designed for Karnataka State Police. Real-time geospatial mapping, predictive policing, network analysis, and intelligent case management across 31 districts.

Event Karnataka State Police Datathon 2026

Version 1.0.0 (Final)

Stack Next.js 16 / TypeScript / Supabase / Tailwind CSS 4

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1. Executive Summary

CrimeSight AI is a comprehensive, AI-powered crime analytics and intelligence platform designed specifically for the Karnataka State Police Datathon 2026. The platform transforms raw First Information Reports (FIRs) and criminal case data into actionable intelligence through eight integrated analytical modules, enabling law enforcement officers, station house officers, and senior leadership to make data-driven decisions in real time.

The system is built on a modern technology stack comprising Next.js 16 with TypeScript, Tailwind CSS 4 with shadcn/ui, Prisma ORM, and Supabase PostgreSQL as the cloud database. It processes a realistic dataset of 500 FIR cases across all 31 districts of Karnataka, with over 8,350 related records spanning suspects, victims, witnesses, evidence, vehicles, arrests, chargesheets, and investigation activities. The platform features a dark-themed, operations-center aesthetic inspired by real-world law enforcement command centers, ensuring both visual impact and functional clarity for datathon judges and police evaluators.

Key capabilities include interactive geospatial crime mapping with district-level drill-down, AI-powered anomaly detection and risk scoring, predictive patrol scheduling based on hourly crime patterns, network intelligence for identifying repeat offenders and cross-district criminal links, cold case revival through automated similarity matching, officer performance leaderboards, voice-based FIR filing through speech-to-text, and a full-screen Command Center mode for presentation scenarios. An always-present AI chat bar provides natural language querying across the entire dataset.

Metric	Value
Total FIR Cases	**500**
Districts Covered	**31**
Total Records (All Tables)	**8,353**
Database Tables	**24**
Officers Tracked	**186**
Analytical Modules	**8**

Table 1: Key Platform Metrics at a Glance

2. Project Context and Problem Statement

2.1 The Challenge

Karnataka State Police handles thousands of criminal cases annually across 31 districts, generating massive volumes of structured data in the form of FIRs, investigation reports, chargesheets, and arrest records. However, this data remains siloed across different police stations and district headquarters, making it extremely difficult for officers and decision-makers to identify patterns, track repeat offenders, allocate resources efficiently, or connect seemingly unrelated cases. Traditional manual analysis is time-consuming, error-prone, and incapable of processing data at the scale required for modern predictive policing.

2.2 The Opportunity

The Karnataka State Police Datathon 2026 presents an opportunity to demonstrate how AI and modern web technologies can transform criminal data into actionable intelligence. CrimeSight AI addresses this by providing a unified, real-time analytics dashboard that combines geospatial visualization, machine learning-driven predictions, natural language processing, and network analysis into a single, intuitive interface designed for law enforcement professionals.

2.3 Target Audience

The platform is designed for three primary user groups within the police hierarchy. First, station-level officers who need to file FIRs, track case progress, and identify leads. Second, district-level SPs and DCPs who require district-wide crime analytics, patrol scheduling, and officer performance monitoring. Third, senior leadership and datathon judges who need a high-level command view with automated insights, anomaly detection, and trend analysis. The Command Center mode is specifically designed for presentation scenarios where the platform must convey its capabilities to evaluators within minutes.

3. Technology Architecture

3.1 Core Stack

The application is built on a modern, production-grade technology stack chosen for its performance, developer experience, and ecosystem maturity. Each component was selected to maximize development velocity for the datathon while maintaining the quality standards expected of a deployable law enforcement application.

Layer	Technology	Purpose
Framework	Next.js 16 (App Router)	Server-side rendering, API routes, file-based routing
Language	TypeScript 5	Type safety across frontend and backend
Styling	Tailwind CSS 4	Utility-first CSS with zero runtime cost
UI Library	shadcn/ui (New York)	Accessible, composable component primitives

Layer	Technology	Purpose
Database	Supabase (PostgreSQL)	Cloud-hosted relational database with real-time
ORM	Prisma 6	Type-safe database queries and schema management
Animations	Framer Motion 12	Declarative page transitions and micro-interactions
Charts	Recharts 2	React-native data visualization library
Maps	D3-geo + SVG	Custom Karnataka district map rendering
Icons	Lucide React	Consistent, tree-shakeable icon set
State	Zustand 5	Lightweight client-side state management
Server State	TanStack Query 5	Caching, refetching, and async state

Table 2: Technology Stack by Layer

3.2 Database Architecture

The data layer uses Supabase PostgreSQL hosted in the Tokyo (ap-northeast-1) region, providing low-latency access and enterprise-grade reliability. The schema was designed from the ground up to model the complete lifecycle of a criminal case in Karnataka, from FIR registration through investigation, arrest, chargesheet filing, and case closure. The Prisma ORM provides a type-safe query interface, eliminating SQL injection risks and ensuring schema consistency.

The database comprises 24 tables organized into four functional groups: Master Data (6 tables for states, districts, units, ranks, designations, and user roles), Crime Taxonomy (6 tables for crime types, categories, acts, sections, evidence types, and vehicle types), Case Management (1 central CaseMaster table with 35 fields), and Case Sub-Entities (11 tables for suspects, victims, witnesses, evidence, property, vehicles, arrests, chargesheets, investigation activities, and case assignments). All String fields use the PostgreSQL Text type for flexibility with variable-length Indian names and addresses.

Table Group	Tables	Records	Key Fields
Master Data	6 (State, District, Unit, Rank, Designation, UserRole)	287	Geographic hierarchy, org structure
Crime Taxonomy	6 (CrimeType, Category, Act, Section, EvidenceType, VehicleType)	110	IPC sections, crime classifications
Case Master	1 (CaseMaster)	500	FIR number, district, priority, status, AI risk score

Table Group	Tables	Records	Key Fields
Suspects	1 (Suspect)	491	Name, arrest status, repeat offender flag
Victims	1 (Victim)	758	Name, injury details, statements
Witnesses	1 (Witness)	782	Name, statement, witness type
Evidence	1 (Evidence)	1,290	Type, forensic status, chain of custody
Property	1 (Property)	763	Type, estimated value, recovery status
Vehicles	1 (Vehicle)	275	Registration, make, model, seizure status
Arrests	1 (ArrestSurrender)	491	Arrest type, datetime, location
Chargesheets	1 (Chargesheet)	373	Court, judge, filing status
Investigation	2 (InvestigationActivity, CaseAssignment)	1,756	Activity type, officer, datetime

Table 3: Database Schema Overview with Record Counts

3.3 Migration Strategy

The project originally used SQLite for local development and was migrated to Supabase PostgreSQL for the production deployment. A dedicated migration script (scripts/migrate-to-supabase.ts) was developed to handle this transition safely. The script exports all data from SQLite, pushes the PostgreSQL schema via Prisma, then batch-imports records using the pg library with 50-row batches and ON CONFLICT DO NOTHING conflict resolution. The entire migration of 8,353 records across 24 tables completed in 14.2 seconds. A rollback script (scripts/rollback-to-sqlite.sh) is maintained as a safety net, along with the original SQLite schema backup at prisma/schema.sqlite.prisma.

4. Feature Specifications

CrimeSight AI delivers its capabilities through eight primary modules, each accessible via a tabbed navigation interface. Every module is designed to operate independently while sharing a unified design language and common data layer. The following sections detail each module's functionality, data sources, user interactions, and API architecture.

4.1 Crime Map (Geospatial Intelligence)

The Crime Map module serves as the default landing view and provides an interactive SVG-based map of all 31 Karnataka districts. Built with D3-geo for geographic projections and path generation, the map renders each district as a clickable polygon colored by crime density. The color gradient uses an emerald-to-amber-to-red scale where green districts have fewer than 10 cases, amber indicates 10-20 cases, and red highlights districts with over 20 cases. Districts with the highest case counts display a pulsing animation to draw attention to crime hotspots.

Clicking a district opens a detailed slide-over panel showing: total cases, active investigations, critical cases, a breakdown by crime type with horizontal progress bars, the top 5 most recent FIRs with priority badges, and a list of police stations in the district. The left sidebar displays a scrollable alert feed of the 15 most recent cases across all districts, each tagged with priority level (Critical, High, Medium, Low) and case status. Above the map, four summary cards display total cases, active investigations, total arrests, and chargesheet count, all with animated counter transitions. The dashboard API (GET /api/dashboard) serves as the single data source, returning aggregated statistics and recent alerts in one request.

4.2 Network Intelligence

The Network Intelligence module identifies criminal relationships and organizational patterns that would be invisible in individual case records. Following a complete redesign from an initial D3 force-directed graph (which proved too complex for non-technical users), the module now uses a clean, card-based layout with four organized sub-tabs: Crime Networks, Repeat Offenders, Linked Cases, and Cross-District activity.

The Crime Networks sub-tab groups suspects by crime type, displaying cards for each of the 10 identified crime clusters (Theft, Robbery, Assault, Cyber Crime, Fraud, Murder, Kidnapping, Narcotics, Burglary, and Vehicle Theft). Each card shows the number of suspects, linked cases, and a progress bar indicating the cluster size relative to the largest. The Repeat Offenders sub-tab displays a searchable grid of 230 flagged offenders with their FIR numbers, crime types, and districts. The Linked Cases sub-tab identifies suspects appearing in multiple cases using name-based grouping across different case_rowid values, presenting mini SVG hub diagrams showing the connected cases. The Cross-District sub-tab highlights suspects operating across multiple districts with visual connection lines between their operating areas. A detail dialog provides expanded profiles with full case lists and crime type breakdowns.

4.3 AI Analysis

The AI Analysis module is the intellectual core of CrimeSight AI, providing two primary capabilities: automated anomaly detection and interactive AI-powered risk scoring. The module features an animated neural network SVG visualization in its idle state, which transitions to a real-time analysis view when the user triggers a scan.

The anomaly detection engine identifies three types of patterns: spikes (sudden increases in crime rates for a specific district or crime type), patterns (recurring temporal or geographic correlations), and outliers (statistically unusual cases that deviate from expected behavior). Each anomaly is tagged with a severity level (Critical, Warning, Info) and includes the district, metric, and timestamp. The risk scoring system assigns AI risk scores to cases based on multiple factors including crime type severity, geographic clustering, temporal patterns, and suspect history. High-risk cases are surfaced with visual indicators and recommended actions. The analysis uses pattern-matching algorithms against the historical dataset to identify statistically significant deviations without requiring external ML model training.

4.4 Crime Trends

The Crime Trends module provides four distinct analytical views rendered through Recharts visualizations. The Crime Type Distribution view displays a horizontal bar chart ranking all crime types by frequency, enabling officers to quickly identify the most prevalent offenses. The District Comparison view presents a grouped bar chart comparing case volumes across districts, sortable by total cases, active cases, or critical cases. The Time-of-Day Analysis view renders a 24-hour line chart showing crime frequency by hour, with reference lines marking the mean and identifying peak hours (identified as hours 1, 17, and 22 in the current dataset). The Monthly Trends view displays an area chart showing case registration patterns over time, enabling seasonal and temporal trend identification.

Each view includes interactive tooltips, color-coded data series, and summary stat cards highlighting key findings such as the most dangerous hour, the highest-crime district, and the fastest-growing crime type. All four API endpoints (/api/trends/crime-types, /api/trends/district-comparison, /api/trends/time-of-day, /api/trends/monthly) return pre-aggregated data for instant rendering without client-side computation.

4.5 Case Management

The Case Management module provides comprehensive FIR lifecycle management with search, filter, detail view, creation, and two AI-powered tools: Case Cracker and Voice FIR. The main view displays a paginated, searchable table of all 500 cases with columns for FIR number, crime type, district, status, priority, and date. Filters allow narrowing by district, crime type, status, and priority. The search supports FIR number, place name, and free-text queries.

The Case Cracker (accessible via a rocket icon on each case row) is an AI-powered analysis tool that scans the entire case database to generate investigative leads for a specific case. It identifies five lead types: repeat suspects (suspects with prior cases in the database), area patterns (geographic clustering near the crime location), vehicle links (vehicles associated with other cases), property matches (stolen property matching other cases), and time patterns (temporal correlations with other crimes). Each lead includes a confidence score (0-100%), a description, linked cases, and a recommended action. The analysis processes all 500 cases in under 2 seconds with an animated progress bar showing the scanning phases.

The Voice FIR feature enables officers to file FIRs by speaking into their device microphone. Using the Web Speech API through a custom useSpeechRecognition hook, the system provides a 3-step workflow: Record (real-time transcription with waveform visualization), Review (editable auto-parsed fields for crime type, district, date, and description), and Submit (generates a unique FIR number and saves to the database via POST /api/fir/voice). The modal supports Hindi and English transcription, with auto-detection of crime types and district names from the spoken text.

4.6 Predictive Patrol

The Predictive Patrol module provides time-slider-based crime hotspot prediction across all 31 districts. The primary interface is a horizontal time slider (0-23 hours) that, when adjusted, recalculates predicted crime levels for each district based on historical temporal patterns in the dataset. The API (/api/patrol/predict) analyzes the distribution of crime occurrence times to generate hourly predictions with risk level classifications (Critical, High, Medium, Low).

The display shows hotspot cards for the top-risk districts at the selected hour, each with predicted case count, risk level badge, and a list of recommended patrol zones. A 24-hour area chart below the slider shows the full daily crime pattern, with a vertical reference line indicating the currently selected hour. A district heatmap table provides a comprehensive view of all 31 districts with color-coded risk levels. The system identifies peak hours (1 AM, 5 PM, 10 PM) and recommends proactive patrol deployment during high-risk windows. This module directly addresses the practical need for resource allocation in law enforcement operations.

4.7 Officer Intelligence

The Officer Intelligence module provides a performance leaderboard and unit analysis system for tracking police officer effectiveness. The API (/api/officers/leaderboard) aggregates data from the CaseMaster, Employee, and Unit tables to calculate per-officer and per-unit performance metrics including total cases handled, cases resolved (chargesheeted), resolution rate, and efficiency score.

The main leaderboard displays 50 officers ranked by resolution efficiency, with gold, silver, and bronze medal icons for the top 3 performers. Each row is expandable to show the officer's full profile including rank, unit assignment, contact information, and a breakdown of cases by status. The unit performance grid below shows aggregate metrics for all 62 police units, displayed as cards with case count, resolution rate, and active investigation count. Filters allow searching by officer name and filtering by district. The module provides transparency into law enforcement performance and helps identify units that may need additional resources or support.

4.8 Cold Case Revival

The Cold Case Revival module applies a 6-factor similarity scoring algorithm to match old unsolved cases with recently active cases, potentially generating new investigative leads. The API (/api/cold-cases/match) identifies cold cases (defined as cases filed more than 90 days ago with no chargesheet and no recent investigation activity) and scores them against active cases based on: crime type match (weight 25%), geographic proximity (20%), temporal pattern similarity (15%), suspect profile overlap (15%), modus operandi indicators (15%), and evidence type correlation (10%).

The current dataset contains 174 identified cold cases, with 20 high-confidence matches generated by the algorithm. Each match card displays the cold case and active case side-by-side with an SVG score ring showing the overall match confidence (0-100%). Match reasons are listed with descriptive icons (e.g., same crime type, nearby district, similar time window). A cold case age badge indicates how long the case has been unsolved (e.g., "180 days cold"). Filters allow narrowing by district, crime type, and minimum match score threshold. Investigation recommendations are generated based on the specific match factors, suggesting which evidence to re-examine or which witnesses to re-interview.

5. Special Features

5.1 Command Center Mode

The Command Center Mode is a full-screen, auto-rotating presentation view designed for demonstrating CrimeSight AI's capabilities to datathon judges and senior leadership. Activated by pressing the F key or clicking the CC button in the header, this mode takes over the entire viewport with a cinematic dark theme, hiding the standard navigation and footer. The view automatically cycles through five panels every 10 seconds: Crime Map overview with district rankings, AI anomaly feed with severity indicators, Top Cases summary with priority badges, Officer Leaderboard top 5, and Network Intelligence summary.

A countdown timer shows seconds until the next panel rotation. A minimap in the corner provides context for the current view. A horizontal scan-line animation and "COMMAND CENTER MODE" badge in the header reinforce the operations-center aesthetic. Pressing F or Escape exits the mode and returns to the standard interface with all previous state preserved. This feature is critical for datathon presentations where time-limited demonstrations must convey maximum information with minimal interaction.

5.2 AI Chat Bar

The AI Chat Bar is a persistent, always-visible component at the bottom of the screen that provides natural language querying across the entire CrimeSight AI dataset. The interface consists of a compact input bar that expands upward into a scrollable chat history when activated. Users can ask questions in plain English such as "How many theft cases in Bangalore Urban?" or "Show me repeat offenders in Mysore" and receive contextual responses powered by the backend LLM integration.

The chat bar features a typing indicator animation during AI response generation, auto-scroll to the latest message, and a collapsible interface that minimizes to a single-line input when not in use. The API endpoint (POST /api/ai/chat) processes the user query against the database schema and generates structured responses. The chat history is maintained in client-side state for the session duration. This feature eliminates the need for users to navigate to specific tabs to find information, making the platform accessible to non-technical users who prefer conversational interfaces.

5.3 Boot Sequence

The application features a cinematic boot sequence that plays on first visit, displaying the CrimeSight AI shield logo, the application name with emerald accent, and a horizontal progress bar that fills from 0% to 100% over 1.3 seconds. The text "LOADING SYSTEMS..." with a percentage counter provides a sense of system initialization. Once complete, the boot screen fades out with a 300ms opacity transition. The boot state is stored in sessionStorage to prevent replay on subsequent page loads within the same session, ensuring the experience is impactful on first impression but does not delay returning users.

6. API Architecture

The backend follows Next.js App Router conventions with all API routes defined under src/app/api/. Each route is a self-contained module that queries the database via Prisma Client and returns JSON responses. The API layer is designed for zero client-side database access, ensuring all data operations go through validated server-side endpoints. The following table documents the complete API surface area.

Endpoint	Method	Purpose
/api/dashboard	GET	Aggregate stats, recent alerts, district summary
/api/cases	GET	Paginated case list with search and filters
/api/cases/[id]	GET	Full case detail with all sub-entities
/api/cases/[id]/crack	GET	Case Cracker AI lead generation
/api/network/repeat-offenders	GET	Repeat offenders, clusters, cross-district
/api/network/graph	GET	Network graph nodes and edges
/api/ai/anomalies	GET	Anomaly detection results
/api/ai/risk-scores	GET	Case risk scoring distribution
/api/ai/chat	POST	Natural language query processing
/api/trends/crime-types	GET	Crime type frequency distribution
/api/trends/district-comparison	GET	District-level case comparison
/api/trends/time-of-day	GET	24-hour crime frequency analysis
/api/trends/monthly	GET	Monthly case registration trends
/api/patrol/predict	GET	Hourly crime prediction per district
/api/officers/leaderboard	GET	Officer and unit performance metrics
/api/cold-cases/match	GET	Cold case similarity matching
/api/fir/voice	POST	Voice FIR submission endpoint
/api/map/stats	GET	District-level map statistics
/api/map/district-detail	GET	Single district detailed view data
/api/districts	GET	All districts list
/api/districts/[id]	GET	Single district detail
/api/master/[type]	GET	Master data by type (ranks, units, etc.)

Table 4: Complete API Endpoint Registry

7. UI/UX Design System

7.1 Visual Identity

CrimeSight AI employs a dark, operations-center-inspired visual design that immediately communicates its law enforcement context. The primary background color is #030712 (near-black with a subtle blue undertone), creating a high-contrast canvas for data visualization. The emerald green accent (#10B981) serves as the primary interactive color, used for the brand identity, active tab indicators, and positive status indicators. Amber (#F59E0B) represents warnings and medium-priority items, while red (#EF4444) marks critical alerts and high-priority cases. Cyan (#06B6D4) is used for informational elements and chargesheet counts. Text uses slate-200 (#E2E8F0) for primary content and slate-500 (#64748B) for muted/secondary text, ensuring readability without eye strain during extended use.

7.2 Interaction Patterns

The interface uses Framer Motion for all transitions and micro-interactions. Tab switches use cross-fade animations (200ms opacity transition) with a spring-physics animated underline that slides between tabs using layoutId for smooth position tracking. Cards and list items stagger in with 70ms delays between each element using a spring easing curve. The boot sequence uses a linear progress bar. The Command Center mode cross-fades between panels. All interactive elements have hover states with color transitions. Modal dialogs use scale-and-fade entrance animations. The live clock in the header updates every second, and a pulsing green dot indicates the system is live.

7.3 Responsive Design

The application is fully responsive across mobile, tablet, and desktop viewports. The header stats bar is hidden on screens smaller than the md breakpoint (768px) to preserve space. Tab labels are always visible but use compact spacing on mobile. The Command Center button text is hidden on small screens, showing only the icon. The crime map and all chart components use ResponsiveContainer from Recharts to adapt to available width. All touch targets maintain a minimum size of 44px for mobile accessibility. The AI Chat Bar auto-expands based on available screen width.

7.4 Component Library

The application uses the complete shadcn/ui component library (New York style variant) for all interactive primitives including buttons, inputs, selects, dialogs, tabs, badges, cards, tooltips, and scroll areas. These components provide built-in accessibility (ARIA attributes, keyboard navigation, screen reader support) and consistent styling through CSS custom properties. Custom components are built on top of this foundation: the AI Chat Bar, Command Center Mode, Case Cracker Modal, Voice FIR Modal, and all tab components use shadcn/ui primitives as their building blocks. The Lucide icon library provides a consistent icon language across all modules, with over 30 unique icons actively used in the interface.

8. Project Structure

The project follows Next.js 16 App Router conventions with a clear separation between components, API routes, database schemas, and utility scripts. The following table summarizes the key directories and their contents.

Directory	Contents	File Count
src/app/	Next.js App Router pages and API routes	24 route files
src/components/crimesight/	All feature components (8 tabs + 4 modals)	12 components
src/components/ui/	shadcn/ui primitive components	50+ components
src/hooks/	Custom React hooks (speech recognition)	1 hook
src/lib/	Utility modules (db client, helpers)	3 files
prisma/	Database schema, migrations, seed data	4 files
scripts/	Migration scripts, shell utilities	4 files
public/	Static assets, GeoJSON data	8 files

Table 5: Project Directory Structure

9. Deployment and Operations

The application is designed for deployment on Vercel with Supabase as the backend database. The current development setup uses a local dev server (bun run dev) with a keepalive wrapper (while true; do bun run dev; done) to handle automatic restarts during development. The production build command (bun run build) generates a standalone Next.js server with static assets copied into the output directory.

Environment configuration is managed through .env files with two database connection strings: DATABASE_URL (for Supabase connection pooler, used during build time) and DIRECT_DATABASE_URL (for direct database connections, used during runtime and migrations). The db.ts module uses dotenv with override:true to force-read environment variables, preventing stale cached values. A Caddy-based gateway handles external port exposure, routing requests through XTransformPort query parameters to appropriate backend services.

The development environment includes several quality assurance tools: ESLint (bun run lint) for code quality, Prisma Studio for database inspection, and automated browser testing via the agent-browser tool for end-to-end verification. The project enforces TypeScript strict mode throughout, with all API routes and components fully typed. The total dependency count is 94 packages (81 runtime, 13 dev), with a focus on tree-shakeable libraries to minimize bundle size.

10. Acceptance Criteria and Success Metrics

The following criteria define the minimum requirements for the CrimeSight AI platform to be considered complete and ready for datathon submission.

ID	Criterion
C-01	All 8 tab modules load and display data correctly without runtime errors
C-02	Interactive Karnataka district map renders with correct color coding by crime density
C-03	AI anomaly detection generates at least 5 anomalies with correct severity tags
C-04	Case Cracker produces leads with confidence scores for any selected case
C-05	Voice FIR records, transcribes, parses, and submits an FIR via microphone input
C-06	Predictive Patrol slider updates hotspot predictions for all 24 hours
C-07	Officer Leaderboard displays at least 50 officers with correct rankings
C-08	Cold Case Revival generates matches with score rings above 0% confidence
C-09	Command Center Mode auto-rotates through all 5 views without manual interaction
C-10	AI Chat Bar accepts natural language queries and returns contextual responses
C-11	All 22 API endpoints return valid JSON with correct data types
C-12	Database contains all 8,353 records across 24 tables with no data loss
C-13	Application renders correctly on mobile (375px), tablet (768px), and desktop (1440px)
C-14	Boot sequence plays once per session and is skipped on subsequent visits
C-15	Dark theme maintains WCAG AA contrast ratios for all text elements

Table 6: Acceptance Criteria Checklist